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Does accumulated physical load in different time windows affect hamstring injuries in elite football players?

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ABSTRACT

Background: This study aims to investigate how accumulated training load over different time windows (7, 14, and 28 days) influences the incidence of hamstring injuries in elite football players.

Methods: A descriptive and longitudinal study was developed in 343 players in three teams (2nd Team, U-19 team, and U-18 team) from the academy of an elite football club in Spanish La Liga that also competed in the Union of European Football Associations Champions League during four seasons, from 2017/18 to 2020/21. External load variables (total distance, high-speed running, accelerations, etc.) for training and competition have been studied considering the volume variables during three different time windows along the different seasons (7 days, 14 days, and 28 days).

Results: Results did not show differences in the shortest time windows (7- and 14-week periods), showing that strength and conditioning coaches should analyse longer periods of time and compare them to previous periods with the same length in order to improve workload management and reduce injury risk.

Conclusions: It is important consider the accumulated load (chronic) in periods of 28 days to avoid an increase that generates a hamstring injury. The control of the training load in blocks of 28-day time windows seems essential to minimize the injury rate in football players.

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External load; workload; high-speed running; soccer; acute:chronic ratio

Introduction

The success of a football team is highly associated with player availability (Ekstrand, 2013) and muscle injuries are the main cause of absence among professional football players (Hägglund et al., 2001). Hamstring tears are the most common injury in football, representing 12% of all professional football injuries (Ekstrand et al., 2011). In this

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regard, a 25-player squad might suffer six hamstring injuries each season, which is equivalent to more than 80 days of football activities lost due to injury (Ekstrand et al., 2011). The economic burdens that injuries cause is an important point to address during the season for professional football clubs (Ekstrand, 2013). Muscle injuries constitute more than one-third of all time-loss injuries and cause more than a quarter of total injury absence in high-level European professional football clubs (Ekstrand et al., 2011).

Training load monitoring analysis is essential to reduce overuse injuries related to training load exposure (Weston, 2018). Electronic performance tracking systems, such as global positioning systems (GPSs), have become the most common device for monitoring training and match load in professional football teams (Cummins et al., 2013; Hader et al., 2019). This technology allows obtaining information about how the players are adapting to the training stimulus and what are the physical demands required during competition (Oliva Lozano et al., 2022). The availability of this information may assist coaches, fitness coaches, and medical staff to minimize the injury risk and to maximize the fitness readiness and performance of players (Rago et al., 2020).

Physical demands related to high-intensity (HI) efforts have increased significantly over the latest seasons in football (Barnes et al., 2014; Zhou et al., 2020). These HI actions are the most relevant and determinant in football results (Sweeting et al., 2017) but these efforts have previously shown a correlation with muscle injury risk in volume, number of efforts, or density (Malone et al., 2018). The injury prevention paradox states that excessive acute loads increase the risk of injury, whereas chronic exposure to higher loads enhances the physical capacities of athletes. It makes them more resilient to injury while also enhancing performance (Gabbett, 2016). Consequently, there has been growing support for the acute:workload ratio (ACWR) as a method of prescribing appropriate training loads (Bowen et al., 2020). This involves the assessment of the absolute one-week training load (acute workload) relative to different week windows of chronic training load (window week average acute training load). The training load index is calculated by indicating whether the individual's acute training load is greater, less than, or equal to the chronic training load he or she has been prepared for (Andrade et al., 2020). Although its effectiveness has been criticized in the scientific literature (Impellizzeri et al., 2021). In this sense, it is necessary to quantify both accumulative and weekly changes in load when the aim is to reduce the injury risk (Gabbett, 2016). Although studies have analysed the relation to injury risk, there is still no consensus on what load variables could be more precise for predicting future injuries (Kupperman & Hertel, 2020). Therefore, the aim of this study was to investigate how accumulated training load over different time windows (7, 14, and 28 days) influences the incidence of hamstring injuries in elite football players.

Materials and methods

Participants and inclusion criteria

A descriptive and longitudinal study was developed in 343 players (Table 1) in three teams (2nd Team, U-19 team, and U-18 team) from the academy of an elite football club in Spanish La Liga that also competed in the Union of European Football Associations (UEFA) Champions League during four seasons from 2017/18 to 2020/21.

Table 1. Descriptive participant data.

Season/Team	Players (n)	Age (years)	Weight (Kg)	Height (cm)
2017/18				
2 Team	33	19.72 ± 2.18	73.78 ± 5.97	179.29 ± 6.94
U-19	18	16.76 ± 0.54	69.32 ± 7.48	1.77.7 ± 8.21
U-18	27	15.41 ± 0.64	67.06 ± 8.59	176.57 ± 6.87
2018/19				
2 Team	29	18.57 ± 1.12	72.17 ± 7.41	179.44 ± 7.24
U-19	22	16.38 ± 0.77	70.51 ± 5.92	179.38 ± 5.99
U-18	29	15.41 ± 0.64	67.60 ± 8.83	177.04 ± 7.02
2019/20				
2 Team	31	18.92 ± 1.16	74.07 ± 7.59	181.02 ± 7.01
U-19	24	16.56 ± 0.71	71.17 ± 7.42	170.77 ± 7.16
U-18	29	17.59 ± 2.26	65.70 ± 8.01	176.57 ± 7.57
2020/21				
2 Team	37	19.65 ± 1.93	73.68 ± 5.63	179.78 ± 5.36
U-19	39	16.70 ± 0.99	70.86 ± 7.33	176.51 ± 6.08
U-18	25	15.58 ± 0.64	69.50 ± 7.78	174.00 ± 5.66

Players (n); age, weight, and height (mean ± standard deviation [SD]).

All players belonging to the squads of one of the three teams of each season were eligible for inclusion. Players who were transferred to other clubs or who finished their contracts due to other reasons before the end of a season were included for as long as they trained and played for the club (Ekstrand et al., 2016). The goalkeepers were excluded from the study. We excluded players who had not started the competitive period of the season with their teams. The study protocol was approved by the Club Medical Committee and the local research ethics committee on Science and Ethics of the Club (nº 2019FCB28). This study conformed to the recommendations of the Declaration of Helsinki.

Procedures

- *Description of structured microcycle training model*

A total of 27,895.41 h of training sessions (TSs), 1,488.13 h of friendly matches (FMs), and 6,550.6 h of official matches (OMs) were analysed. Data were collected from U-18 (949 TSs, 67 FMs, and 220 OMs), U-19 (1,095 TSs, 71 FMs, and 226 OMs), and 2nd Team (1,145 TSs, 47 FMs, and 232 OMs). Matches and TS data with the national teams were excluded.

All three teams shared the same training methodology and game model. The training methodology was the structured microcycle (Martín-García et al., 2018), in which football content structuration was usually cyclical but the external load oscillated based on the objectives of each seasonal phase. The first day after the match, the training load was focused on recovery training for the players who played >60 min the day before or a performance compensatory training to try to achieve the training load missed by not having played the match or having participated less than 60 min. After a rest day, the training load was progressively increased up to three days before the next match, progressively decreasing the load until the day of the next match (Guitart et al., 2022).

- *Quantifying workload*

External load was quantified using GPS data collected from all on-pitch TSs and matches during all season phases. Each player was assigned their own specific device in order to avoid potential inter-unit reliability error. GPS data were collected using the WIMU PRO™ device (RealtrackSystems S.L., Almería, Spain). Intra- and inter-unit reliability was acceptable (the intra-class correlation coefficient value was 0.65 for the x coordinate and 0.85 for the y coordinate) for the systems analysed (Bastida-Castillo et al., 2019). The data collected were analysed using the SPRO™ software (v982, v983, and v984; RealtrackSystems S.L., Almería, Spain) which exports the data in raw format. The GPS devices were calibrated and data consistency was ensured according to previous investigations (Gimenez et al., 2020).

The GPS variables collected for this research are shown in Table 2. External load variables for training and competition have been studied considering the volume variables. The speed thresholds used have been established in previous studies (Miñano-Espin et al., 2017).

- *Muscle injury recording*

This study is based on consensus guidelines for the definitions and data collection procedures for football injury studies described by the UEFA (Fuller et al., 2006). Injuries had to occur during training or match and caused the player to be absent from at least the following TS or match. Injury was defined, using a time-loss definition, as any physical complaint sustained by a player that results from a football match or football training and led to the player being unable to take full part in future football training or match play (Fuller et al., 2006). However, in this research, medical attention is considered muscle injuries of grade 0, according to the British Athletics Muscle Injury Classification, that do not force the player to be away from competition.

Injuries were classified using the OSICS_10 coding system (Orchard Sports Injury Classification System) and muscle injuries were indicated by the code OSICS -M–. In this study, we analysed muscle injuries of the lower limbs in which hamstrings were affected in both myotendinous injury and muscle fibres. All diagnoses were made by the same medical staff (the medical staff was the same during the seasons analysed for all teams).

Table 2. Description of GPS Variables and Speed Zones.

Variable	Unit	Zone
Total distance (TD)	m	Total distance covered including walking, jogging, high-speed running, and sprinting
Medium-intensity running (MIR)	m	Distance covered between 18 and 20.99 km/h
High-intensity running (HIR)	m	Distance covered between 21 and 23.99 km/h
High-speed running distance (HSR)	m	Distance covered above 21 km/h
Sprint distance (SpD)	m	Distance covered above 24 km/h
Sprint count (Sp)	n	Total actions above 24 km/h
High metabolic load (HML) distance	m	Distance covered above 25 W/kg^2
Acceleration distance (HI_{ACC} Dist.)	m	Total distance covered accelerating above 3 m/s^2
Deceleration distance (HI_{DCC} Dist.)	m	Total distance covered decelerating below −3 m/s^2

- *Time windows*

Three different time windows were set (7 days, 14 days, and 28 days). For each time window, accumulative workload and ACWR were calculated. Each injury was analysed using two-time blocks: the injury block and the pre-injury block. The injury block represents the days immediately preceding the injury, while the pre-injury block represents the days before the injury block. For example, in a 7-day time window, the injury block includes the 7 days leading up to the injury, and the pre-injury block includes the 7 days before the injury block.

Statistical analysis

The normality distribution and the homogeneity of variance were confirmed with the Kolmogorov – Smirnov test and Levene's test, respectively. The injury ID was the observation level. The performance variables (physical demands) were compared between injury block and the pre-injury block for the different injury time windows (7 days, 14 days, and 28 days). Because some players presented more than one injury, linear mixed models were used to include the player as a random effect to control the repeated measurement. Furthermore, the same performance variables were compared between the different types of injuries (injury or medical attention) for the injury block in the different time windows (7 days, 14 days, 28 days and rolling average ACWR 7:28) using independent sample t-test. The confidence level was established at 95%, considering values of $p < 0.05$ to be statistically significant. Furthermore, differences were also studied using the standardized effect size (ES) differences using Cohen's coefficient. The ES was interpreted as follows: <0.2 = negligible; 0.2–0.6 = small; 0.6–1.2 = moderate; > 1.2 = large. IBM SPSS Statistics version 28.0 software for Windows (SPSS Inc., IL, USA) was used.

Results

The results revealed differences in the 28-day time window for all analysed variables ($p < 0.05$; Table 3) in the injured players. The players showed higher external load in the injury block in comparison to the pre-injury block in terms of TD (+13605.61 m; CI95%: 5246.03 to 21,965.19; ES: 0.50), HSR distance (+725.41 m; CI95%: 251.47 to 1199.36; ES: 0.48), HI_{ACC} distance (+119.42 m; CI95%: 45.69 to 193.15; ES: 0.49), HI_{DCC} distance (+133.99 m; CI95%: 59.44 to 208.55; ES: 0.57), number of sprints (+18.81; CI95%: 6.53 to 31.09; ES: 0.48), and HML distance (+2689.54 m; CI95%: 1075.33 to 4303.75; ES: 0.55). The other time windows did not evidence any significant difference in the parameters included in the study ($p > 0.05$).

Regarding the distance covered in the different speed zones, the distance was also higher in the 28-day time window in the injury block for zone 18–21 km/h (+581.83 m; CI95%: 206.95 to 956.71; ES: 0.51), zone 21–24 km/h (+333.08 m; CI95%: 126.44 to 539.72; ES: 0.52), and zone >24 km/h (+204.40 m; CI95%: 22.08 to 386.72; ES: 0.37; $p < 0.05$; Figure 1) when compared to the pre-injury block.

According to the type of injury, the analysis of variance revealed a higher load in all the time windows before the injury of the players with a hamstring muscle injury for HSR distance ($ES_{7\ days} = 0.66$; $ES_{14\ days} = 0.61$; $ES_{28\ days} = 0.72$), HI_{ACC} distance ($ES_{7\ days} = 0.34$;

Table 3. External load of last and previous 7-day, 14-day, and 28-day time windows of hamstring-injured players.

		Pre-injury block			Injury block			p value
Total distance (m)	7 days	23158.38	±	8790.31	23812.67	±	7625.76	0.58
	14 days	41060.58	±	14773.33	43193.92	±	12927.39	0.29
	28 days	67460.97	±	29918.73	81066.57	±	24183.49	0.00
HSR distance (m)	7 days	1093.34	±	607.72	1151.94	±	598.46	0.51
	14 days	1914.40	±	804.06	2031.18	±	905.63	0.36
	28 days	3068.65	±	1557.62	3794.06	±	1467.63	0.00
HI_{ACC} distance (m)	7 days	210.75	±	78.38	212.56	±	62.89	0.87
	14 days	382.07	±	144.46	389.57	±	107.41	0.70
	28 days	622.67	±	264.11	742.09	±	218.51	0.00
HI_{DCC} distance (m)	7 days	208.98	±	79.98	212.44	±	72.63	0.77
	14 days	370.44	±	142.16	385.92	±	116.08	0.42
	28 days	592.91	±	249.00	726.90	±	222.89	0.00
Number of sprints (n)	7 days	26.70	±	15.88	28.52	±	16.27	0.44
	14 days	45.50	±	21.47	50.04	±	24.28	0.17
	28 days	73.07	±	39.31	91.88	±	39.41	0.00
HML distance (m)	7 days	4194.31	±	1744.29	4417.28	±	1574.12	0.37
	14 days	7452.44	±	2766.77	7885.60	±	2544.24	0.27
	28 days	12073.08	±	5166.78	14762.63	±	4593.21	0.00

Data are mean ± SD; * Significant differences between last and previous time windows ($p<0.05$); HSR: high-speed running; HI_{ACC}: high-intensity acceleration; HI_{DCC}: high-intensity deceleration; HML: high metabolic load.

$ES_{14\ days} = 0.47$; $ES_{28\ days} = 0.54$), and number of sprints ($ES_{7\ days} = 0.93$; $ES_{14\ days} = 0.81$; $ES_{28\ days} = 0.91$), in comparison to the players with a muscle medical attention (Table 4; $p < 0.05$). The TD, HI_{DCC} distance, and HML distance accumulated for the players in the 7-day time window before the injury were not significantly different between the types of injuries ($p > 0.05$), but the 14-day and 28-day time windows before the injury showed a significantly higher load in these variables in the players with a hamstring muscle injury ($p < 0.05$). Distances in zone 18–21 km/h ($ES_{7\ days} = 0.39$; $ES_{14\ days} = 0.43$; $ES_{28\ days} = 0.58$), zone 21–24 km/h ($ES_{7\ days} = 0.55$; $ES_{14\ days} = 0.55$; $ES_{28\ days} = 0.65$), and zone >24 km/h ($ES_{7\ days} = 0.73$; $ES_{14\ days} = 0.55$; $ES_{28\ days} = 0.70$) are also significantly higher for muscle injuries (Figure 2). The ACWR was not different between the players with a hamstring tear and the players with a medical attention in any of the analysed variables ($p > 0.05$).

Discussion

The aim of this study was to investigate the relation between accumulated physical load variables and ACWRs in different time windows with hamstring injuries in elite football. This investigation was based on previous studies that suggested that lower limb non-contact muscle injuries were higher when the external load increased in the previous week (Carey et al., 2017; Thornton et al., 2017). The main finding of this study was that the hamstring-injured players showed higher external load differences in the injury block in comparison to the pre-injury block for a 28-day time window in all external load variables studied. Thereby, the control of the training load in blocks of 28-day time windows seems essential to minimize the injury rate in football players.

There is a large number of studies investigating risk factors for injury regarding training load behaviour and many of them demonstrated that greater amounts of very high-velocity running distance and TD are associated with an increased risk of lower-body soft-tissue injury (Gabbett & Ullah, 2012; Soligard et al., 2016). High accumulative loads and

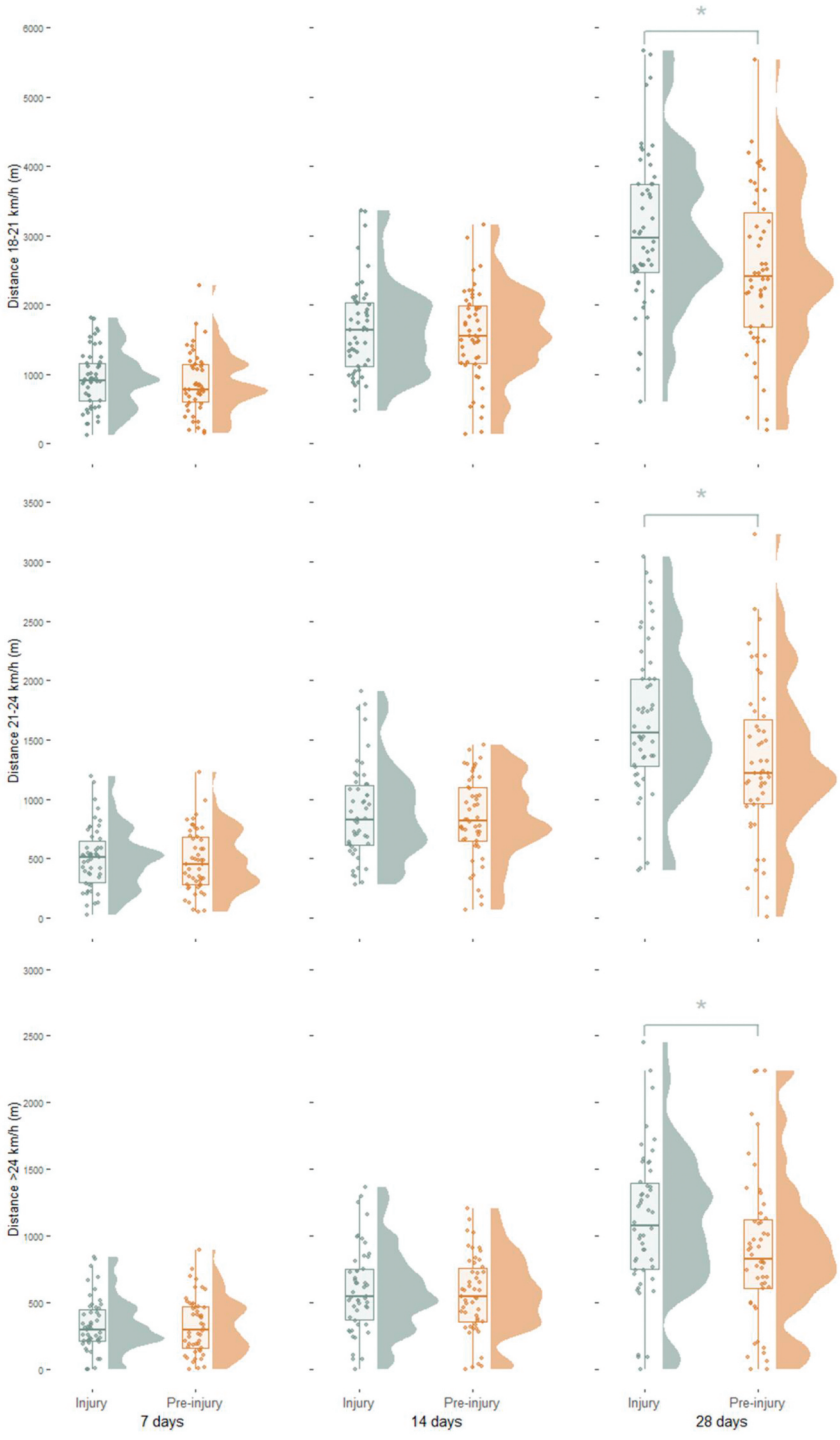

Figure 1. Violin plot of the distances covered in the different speed zones for the 7-day, 14-day, and 28-day time windows in the pre-injury and injury block. * Significant differences between last and previous time windows ($p<0.05$).

Table 4. External load of last 7-day, 14-day, 28-day time windows in football players before a hamstring muscle medical attention or a hamstring injury.

		Hamstring muscle medical attention			Hamstring muscle injury			p value
Total distance (m)	7 days	23113.57	±	6584.54	26026.47	±	10304.00	0.25
	14 days	41404.26	±	10261.18	48861.17	±	18551.73	0.08
	28 days	76599.80	±	18073.54	95211.36	±	34918.05	0.02
	ACR	1.04	±	0.15	1.00	±	0.12	0.34
HSR (m)	7 days	1016.16	±	504.58	1581.93	±	688.20	0.00
	14 days	1818.75	±	738.21	2703.88	±	1082.08	0.00
	28 days	3367.88	±	1156.90	5143.64	±	1574.32	0.00
	ACR	1.00	±	0.36	1.11	±	0.30	0.36
HI_{ACC} distance (m)	7 days	201.97	±	61.35	246.08	±	57.82	0.03
	14 days	363.34	±	97.32	472.62	±	98.13	0.00
	28 days	682.58	±	178.41	930.56	±	233.76	0.00
	ACR	1.02	±	0.18	0.97	±	0.12	0.32
HI_{DCC} distance (m)	7 days	207.13	±	75.85	229.25	±	61.15	0.36
	14 days	368.16	±	113.31	442.18	±	110.88	0.05
	28 days	683.68	±	208.25	863.76	±	220.31	0.01
	ACR	1.04	±	0.19	0.98	±	0.14	0.31
Number of sprints (n)	7 days	24.08	±	13.84	42.58	±	15.85	0.00
	14 days	43.32	±	19.86	71.33	±	25.49	0.00
	28 days	79.34	±	31.90	131.58	±	35.14	0.00
	ACR	0.98	±	0.41	1.17	±	0.31	0.15
HML distance (m)	7 days	4197.94	±	1434.18	5111.86	±	1851.11	0.08
	14 days	7434.48	±	2169.90	9314.16	±	3173.02	0.02
	28 days	13680.88	±	3717.17	18188.16	±	5537.15	0.00
	ACR	1.06	±	0.20	1.03	±	0.15	0.68

Data are mean ± SD; * Significant differences between hamstring muscle medical attention and hamstring muscle injury ($p<0.05$); HSR: high-speed running; HI_{ACC}: high-intensity acceleration; HI_{DCC}: high-intensity deceleration; HML: high metabolic load.

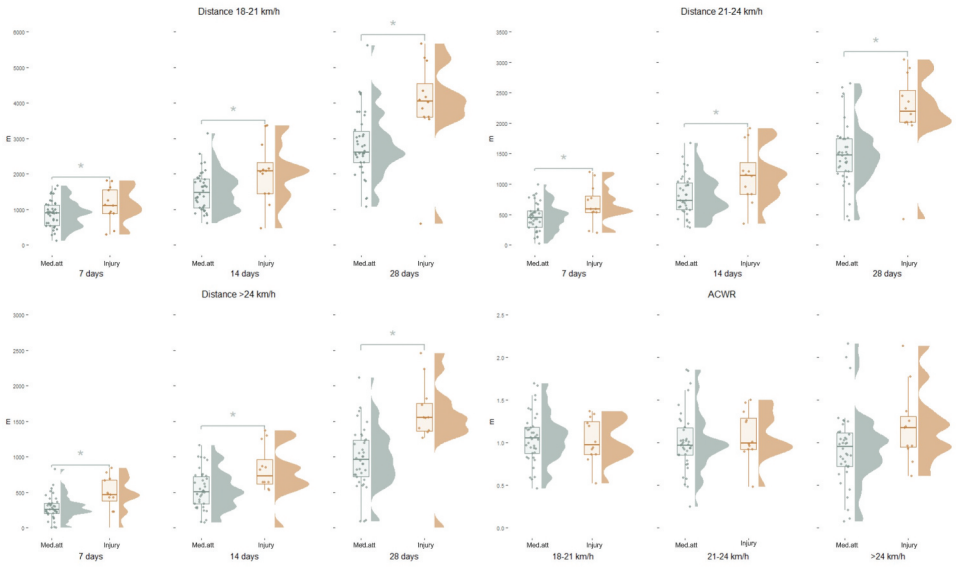

Figure 2. Violin plot of the distances covered and ACWR in the different speed zones for the 7-day, 14-day, and 28-day time windows in hamstring-injured players. * Significant differences between last and previous time windows ($p < 0.05$).

TDs have been identified as risk factors for injury in football. Jaspers et al. (2018) reported an increased injury risk for players that reached higher cumulative loads in 1-, 2-, and 3-week loads in TD in elite football players. In this sense, Bowen et al. (2017) found that high accumulated TD, HSR distance, number of accelerations, and total load over 4 weeks were associated with a significant increase in injury risk for male youth football players. Therefore, previous research determined that large weekly changes in HSR distance and sprint distance increase the injury risk; however, in this study, no significant differences were found in any variables between the injury and pre-injury blocks for 7-day time windows (Malone et al., 2018). Similar analyses were developed by Ehrmann et al. (2016) who compared the injury block for two-time windows (1 and 4 weeks) with the season average load. No significant differences were found for the variables TD, HSR, and very high-intensity running (sprint distance), whereas in this study we showed significant differences in these variables in the 28-day time window. Thus, it might be more important to analyse the previous training load performed than the seasonal average.

Duhig et al. (2019) found that large and rapid increases in individualized sprint distance should be avoided to prevent hamstring injuries in Australian football. This finding is in line with our results and probably indicates the relevance of the use of relative and individualized thresholds to examine the relationship with injury risk (Buchheit, 2017), not just high-speed distance but other external load variables as Jaspers et al. (2018) found with similar results as our study regarding TD, sprint distance, HIACC distance, and HIDCC distance.

Our purpose was to analyse different time windows to improve these findings and draw further conclusions. We did not find differences in the shortest time windows (7- and 14-week periods) showing that strength and conditioning coaches should analyse longer time periods and compare them to previous periods with the same length in order to improve training load management and reduce injury risk.

Previous studies (Andrade et al., 2020; Bowen et al., 2020) have already analysed the relation between the ACWR and the injury risk in multiple sports, including football. In this study, the method to calculate the ACWR was the rolling average 7:28. The same method was used in previous studies to relate the injury risk to the ACWR in elite football players (Bowen et al., 2020; Duhig et al., 2019). The findings of these studies suggest that higher ACWR values are likely to cause injury risk. However, those studies did not analyse the severity of the injury as done in this research. Our findings suggest that the injury severity is not determined by the values of the ACWR because no differences were found when comparing injury and medical attention. Additionally, it is important to consider why shorter time windows (7 and 14 days) did not show significant differences compared to the 28-day window. One possible explanation is the individual variability among players, which can influence how they respond to different training loads over shorter periods. Training periodization and match congestion may also play a crucial role. During periods of high match congestion, players may not have sufficient time to recover, potentially masking the effects of accumulated training loads in shorter time windows (Garcia-Unanue et al., 2024). Therefore, longer time windows (28 days) provide a more comprehensive and accurate view of the accumulated load and its relationship with hamstring injuries.

Finally, our findings show a significantly higher load exposure in hamstring muscle injury than medical attention in all external load variables studied. We could conclude that

a high external load exposure during a given time period would influence not only the likelihood of increasing the risk of muscle injury but also its severity. Severe injuries remove players from TSs and matches for lengthy durations and reduce the team's performance (Hägglund et al., 2001); however, few studies have investigated the relationship between training load and injury severity (Andrade et al., 2020; Bowen et al., 2020). Previous research found no accumulated weekly loads for TD, HSR, or sprint distance associated with the severity of injury (Enright et al., 2020). This finding suggests that training load distance-based data have no associative value for the number of days a player will miss following injury. Even though the cited study did not find any association, we found differences in all external load variables regarding the injury severity. Hamstring injuries showed a significantly higher load when compared to medical attention injuries.

The finding that hamstring injuries had higher external loads than medical attention injuries can be explained by several physiological mechanisms. Injuries typically occur due to acute, high-intensity activities that place significant stress on the muscle fibres, leading to sudden damage. These activities often involve rapid acceleration, deceleration, or changes in direction, which generate high external loads (Kalkhoven et al., 2021; Verstappen et al., 2021). In contrast, medical attention injuries are usually the result of repetitive, lower-intensity activities that accumulate over time, leading to gradual wear and tear on the muscle fibres. The difference in the nature of these activities explain why hamstring injuries are associated with higher external loads (Kalkhoven et al., 2021).

Additionally, individual variability in muscle strength, flexibility, and fatigue resistance may influence the susceptibility to injuries under high external loads. Training periodization and match congestion may also contribute, as players might be exposed to higher loads during peak periods of competition, increasing the risk of hamstring injuries (Gabbett & Oetter, 2025).

This study is not without limitations. Firstly, the study considers different teams in different seasons so the workload and physical load may vary (although medical services and performance coaches remained stable). Additionally, variations in training loads between teams and different coaching styles could influence injury risk, which were not controlled for in this study. However, being an ecological study developed in a real context outside the laboratory, it was impossible to incorporate a control group within each of the analysed teams. Finally, only hamstring injuries were considered, excluding any other type of injury.

Practical applications

The results of this study can be useful to balance the distribution of workloads throughout the competitive microcycle to reduce the injury incidence in football players. It is important that physical coaches consider the accumulated load (chronic) in periods of 28 days to avoid an increase that generates a hamstring injury. The control of the training load in blocks of 28-day time windows seems essential to minimize the injury rate in football players.

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Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request. Due to ethical and privacy concerns, the data cannot be made publicly available. Anonymized datasets can be shared with qualified researchers under a data use agreement approved by the institutional ethics committee.

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